温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Review for Midterm

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Midterm Review – Outline

Topics to Review

- Introduction to Business Analytics
- Python Basics, Pandas, Jupyter Notebook
- Descriptive Statistics
- Data Visualization
- Assignment 1&2

Today's Class

- Question-based review to check your understanding
- Recap key concepts and Python functions learned so far
- Get a sense of what to expect in the midterm exam

Introduction to BA

Type	Definition / Purpose	Typical Questions Asked
Descriptive Analysis	Summarizes and interprets *past data* to understand what has happened.	*What happened?* *What are the key patterns or trends?*
Predictive Analysis	Uses *historical data and models* to forecast future outcomes.	*What is likely to happen?* *What factors influence future results?*
Prescriptive Analysis	Recommends *actions or strategies* based on predictive insights.	*What should we do?* *How can we achieve the best outcome?*

- **Categorical (nominal):** classified into distinct categories (no order)
- **Ordinal:** ordered or ranked categories (relative order, no exact distance)
- **Interval:** ordinal, constant differences between observations, arbitrary zero points
- **Ratio:** continuous, equal intervals, a natural zero

Python, Pandas, Jupyter Notebook

- Python Basics
  - What is the difference between `=` and `==`?
  - What if an error occur?
    - SyntaxError, NameError, TypeError, ValueError, KeyError, FileNotFoundError
    - Read the last line of the traceback first — it shows the actual cause of the error.
- Working in Jupyter Notebook
  - How can you view help or documentation for an object or function? (Hint: use `?`, `??`, or `help()`)
  - How can you format text in a Markdown cell? (e.g., bold, italic, headers, bullet points)
- Pandas Operations
  - How to select specific columns or rows that meet a condition?(e.g., `data[data["age"] > 30]`)
  - How to compute descriptive statistics by group? (e.g., using `groupby()` with `.mean()` or `.describe()`)

Descriptive Statistics

- How to Calculate Key Statistics
  - Mean, Median, Mode
  - Variance, Standard Deviation
  - Skewness, Kurtosis
  - Covariance, Correlation Coefficient
  - 👉 Functions: `df.describe(), df.agg(), df.head()`
- Conceptual Understanding
  - When is the median more representative than the mean?
  - What do skewness and kurtosis tell us about data shape?
  - What do covariance and correlation coefficient indicate about relationships between variables?
- Detecting Outliers
  - Z-score rule: values with |z| > 3 may be outliers
  - IQR rule: values outside $Q1-1.5\times IQR$, $Q3+1.5\times IQR$ are potential outliers
- Hands-on Practice
  - Revisit the *Descriptive Statistics Demo* — compute and interpret results on your own

Data Visualization

- Chart Selection
  - How do you choose the **right chart type** based on your analysis purpose?
    - Comparison: bar / column chart
    - Distribution: histogram, boxplot
    - Trend over time: line chart
    - Relationship: scatter plot
- Creating Charts in Python
  - How to draw different charts (e.g., histogram, bar, boxplot, line, scatter)? (Hint: use `plt.plot(), plt.bar(), plt.hist(), plt.boxplot(), plt.scatter()`)
- Customizing Charts
  - How to set **x-axis** and **y-axis labels**? → `plt.xlabel(), plt.ylabel()`
  - How to set a **title** for the chart? → `plt.title()`
- Hands-on Practice
  - Revisit the *Line Chart Demo* — modify labels, title, and style to improve readability.

Assignment 1&2

- Understand the Question
  - Read carefully — what is being asked:
    - concept, calculation, or interpretation?
- Write Clear, Logical, and Concise Answers
  - Avoid unnecessary code or lengthy explanations.
- Practice Tasks
  - Compute descriptive statistics **by group**

    → e.g., `avg_income = data.groupby("gender")["age"].mean()`
  - Draw a **bar chart** to compare different groups

    → e.g., `plt.bar(avg_age.index, avg_age.values)`

Think:

- What is "avg_age"? A Series. What is a Series?
- What is "avg_age.index"? What is "avg_age.values"?

```
df
```

	name	age	final_score	bonus_score	pass	gender
0	Zhangsan	20	78	83	True	male
1	Lisi	19	89	94	True	female
2	Wangwu	21	55	60	False	male
3	Zhaoliu	20	61	66	True	female

```
avg_age = df.groupby('gender')['age'].mean()
avg_age
```

```
gender
female    19.5
male      20.5
Name: age, dtype: float64
```

```
avg_age.index
```

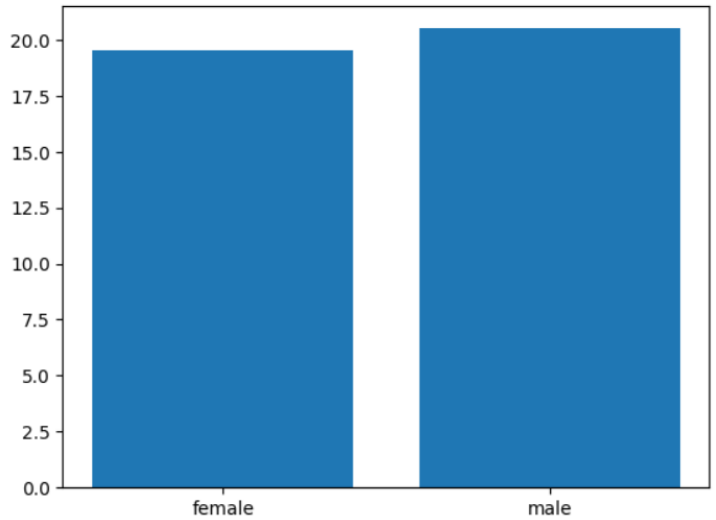
```
Index(['female', 'male'], dtype='object', name='gender')
```

```
avg_age.values
```

```
array([19.5, 20.5])
```

```
plt.bar(avg_age.index, avg_age.values)
```

```
<BarContainer object of 2 artists>
```



Midterm Exam – Logistics & Tips

- When & Where
  - Time: 20:30–21:30 (1 hour, may extend if needed)
  - Date: Tue, Oct 21
  - Venue: TBA
- Allowed Materials
  - One A4 cheat sheet (hard copy).
  - **No AI tools, no web search, no slides, no other materials**.
- Technology
  - Bring your own laptop and ensure Jupyter Notebook runs correctly in advance.
- Files Provided
  - Instructions: `MGS3101_Midterm.ipynb`
  - Dataset: `xxx.csv`
- Tips for the Midterm
  - Read each question carefully (e.g., "**compute and explain**").
  - **Comment briefly** on your output (2-3 sentences).
  - **Save** your work and **double-check** before submitting.